

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Designed and developed **AI-powered agents** using multiple **LLMs** (**OpenAI**, **Claude**, **Llama**, **Gemini**) with the **OpenAI Agents SDK**, **MCP** tool-use, and **prompt engineering** best practices.
- Built **RAG** pipelines with **vector databases** (**Pinecone**) enabling systems that understand **natural language queries** and automate complex workflows.
- Used **CrewAI** and **AutoGen** alongside **LangGraph** as multi-agent orchestration frameworks, handling millions of daily transactions at **99%+ uptime**.
- Built a deployment-gated **model evaluation** framework with automated **data processing** and regression checks, blocking **15%** of failed releases before production.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Trained and **fine-tuned** deep learning models (**PyTorch**) on multi-GPU clusters with **quantization** for efficient inference.
- Improved retrieval recall **61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse retrieval** and re-ranking.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated **25+ production** services in **C++** and **Java** for **1M+ users**, applying strong **data structures and algorithms** foundations, sustaining **99.9% uptime** over 4 years.

PROJECTS

LLaMA 2 Fine-Tuning Pipeline

github.com/ganeshhgupta/llama2-finance-finetuning

- Fine-tuned LLaMA 2** with **LoRA** and **PEFT** for domain-specific document understanding, built from scratch with systematic hyperparameter optimization.

LangFetch

tinyurl.com/LangFetch

- LangGraph** agent with **Pinecone vector** retrieval and hybrid re-ranking; decomposes natural-language questions into dependency-ordered **SQL** subqueries via **FastAPI**.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent** system automating production incident diagnosis via **tool-use** chains; routes alerts to specialized sub-agents and synthesizes root cause reports.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, C++, Java, TypeScript, SQL

Agentic AI: OpenAI Agents SDK, MCP, LangChain, LangGraph, CrewAI, AutoGen, Tool-Use/Function Calling, Prompt Engineering

LLMs & Fine-Tuning: OpenAI, Claude, Llama, Gemini, LoRA, PEFT, Model Fine-Tuning, Quantization

RAG & Retrieval: RAG, Vector Databases (Pinecone), Hybrid Dense-Sparse Retrieval, Re-Ranking

ML Engineering: Data Processing, Model Evaluation, Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks

Cloud: AWS, GCP, Docker, Kubernetes, PostgreSQL